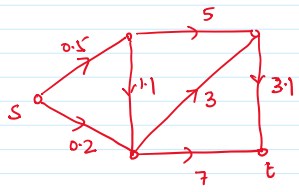
12) | Lecture 22-23 |Single Source Shortest Path Algorithm |

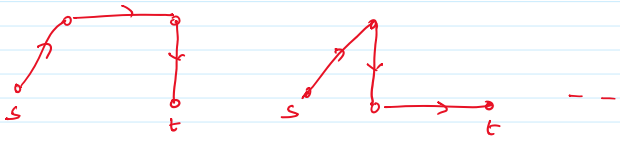
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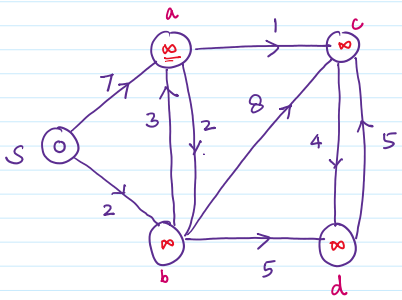
$G = (V, E)$ directed graph

$l : E \rightarrow \mathbb{R}^+$



Aim is to find shortest path from s to t



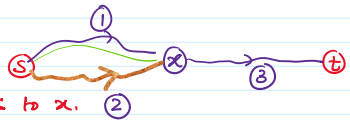


Naive algo is to list down all paths from s to t and report min one.

This could be of exponential time complexity - # of paths can be exponential.

Property of shortest path

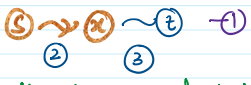
Claim Consider the shortest path from (s) to (t) and (x) is a vertex on the path. Then path from (s) ⇝ (x) is shortest path from s to x.



Proof Let (s) ⇝ (x) is shortest path from s to x (Assume contrary)

$$wt(s \rightsquigarrow x)_{\text{path 2}} < wt(s \rightsquigarrow x)_{\text{path 1}}$$

Consider path from s to t which is



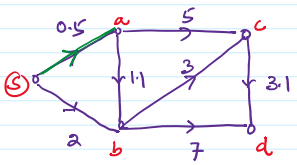
$\rightarrow (1)$

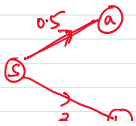
Weight of path given in Eq (1) is smaller than original shortest path

Contradiction!

The claim is true for all vertices (x) sitting on the path.

Problem Stmt: We want to compute shortest path from (s) to remaining vertex in graph.

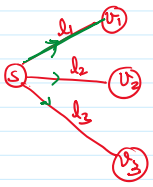




$d[v_i]$:= length of shortest path from s to v_i



$d[v_i]$:= length of shortest path from s to v_i



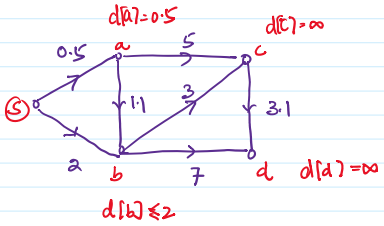
$l_1 < l_2 < l_3$

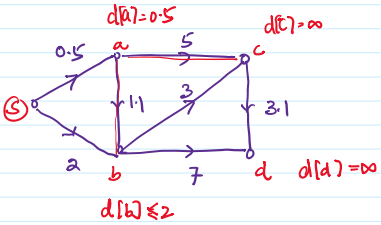
$d[v_1] = l_1$] exact

$d[v_2] \le l_2$] upper bound

$d[v_3] \le l_3$

$d[v_i] = \infty$



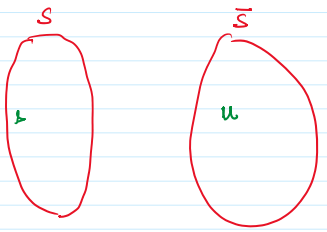


$d[a] + wt(a,b)$

$0.5 + 1.1 < d[b]$

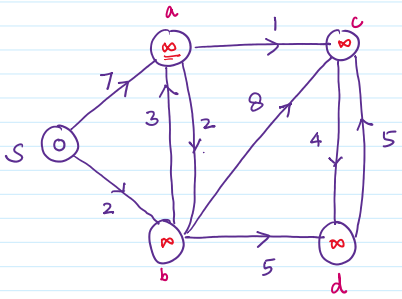
ALG:

In every step of algo. we find the correct shortest path of one more vertex and put to set S



Set of vertices to which we know the correct shortest path from s

We have not found shortest path from s But we know an upper Bound to it



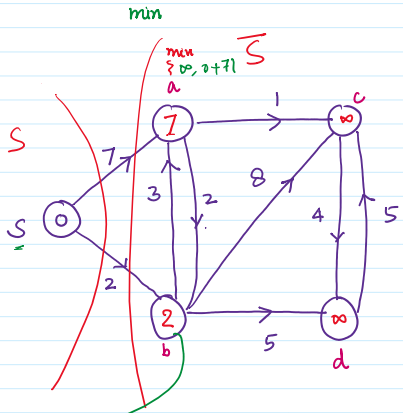
$d[v]$

$\forall w \in V, (u,w) \in E$

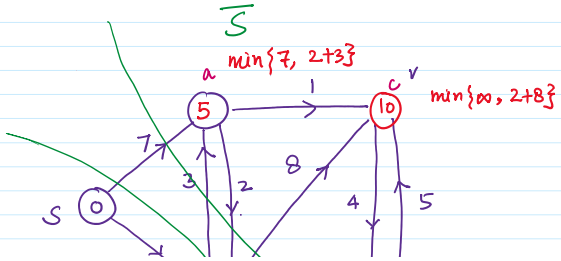
$\min \{ d[w], d[u] + wt(u,w) \}$

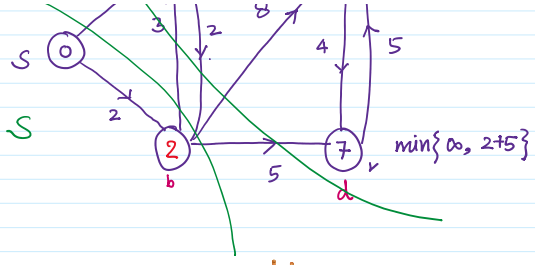
current wt

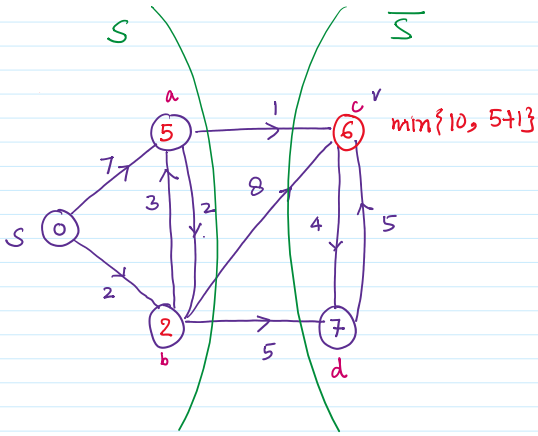
another shortest path from s to w via u

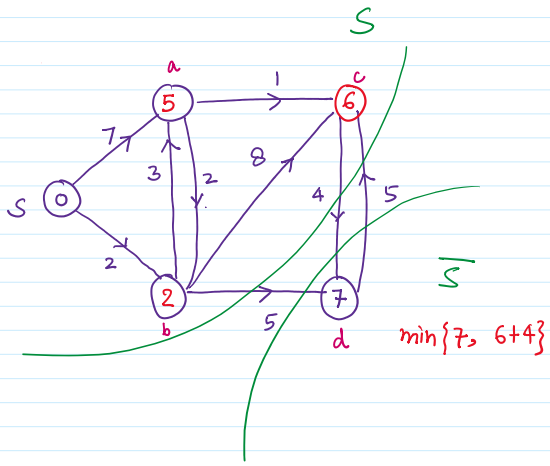


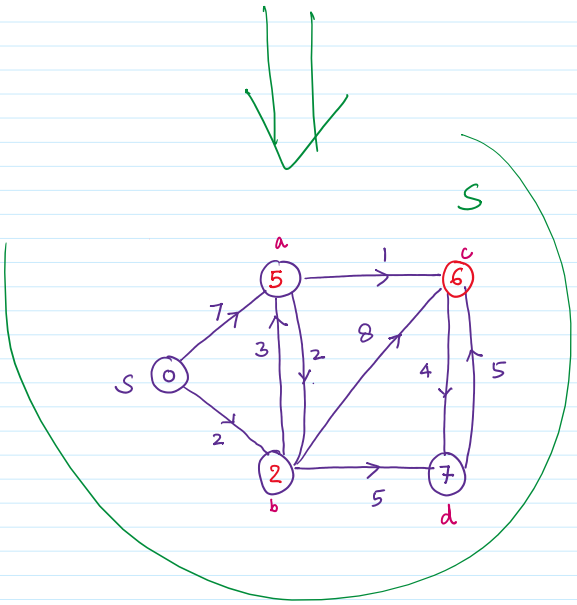
$d[u]$:= length of shortest path from s to u

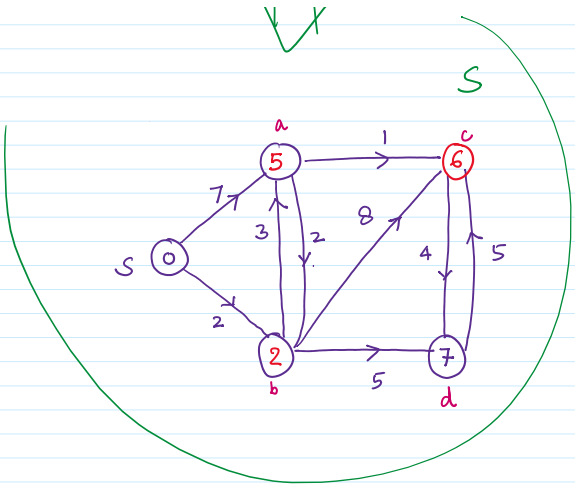



S
S
0
2
3
2
8
4
5
2
b
5
7
d
min{∞, 2+5}

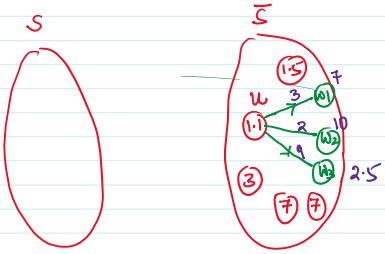

S
S
a
5
1
c
6
min{10, 5+1}
7
3
2
8
4
5
S
0
2
2
b
5
7
d


S
a
5
1
c
6
7
3
2
8
4
5
S
0
2
2
b
5
7
d
S
min{7, 6+4}

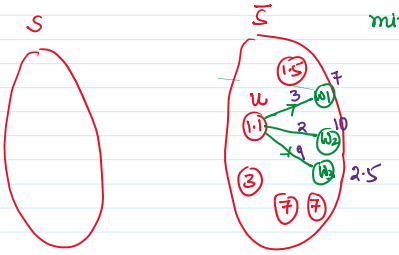

S
a
5
1
c
6
7
3
2
8
4
5
S
0
2
2
b
5
7
d



Prims



SSSP

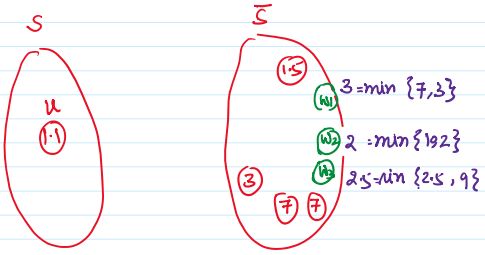


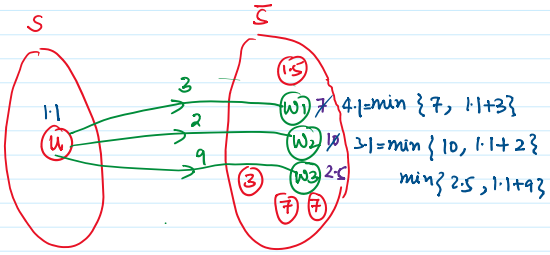
$\forall w \in V, (u,w) \in E$

$\min \{ d[w],\ d[u] + wt(u,w) \}$

current wt

another shortest path from s to w via u





Pseudocode

$\forall v \in V,\ d[v] = \infty,$ H.insert(v, d[v])

$d[s] \leftarrow 0$

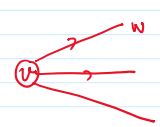
while !H.empty() do

  v = H.findMin() /* vertex with min label] ①

  for all w adjacent to v do {] ②

    ③ [d[w] = min{ d[w], d[v] + wt(v,w) }

    H.decreasepriority(w, d[w]) /* updating label of w in heap



(3) H. decreasepriority (w, d[w]) /* updating label of w in Heap

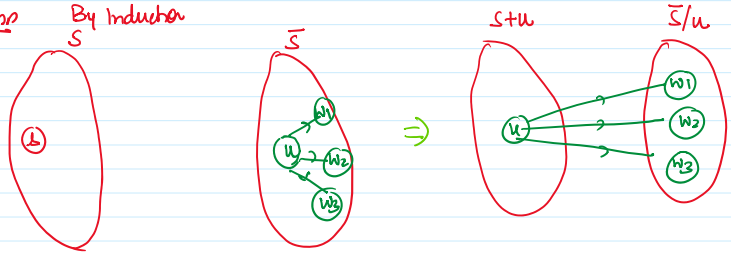
Overall running time $= \sum_{v \in V} (\log n + \deg(v) \log n)$ $\sum_{v \in V} \deg(v) = 2m$

$= O(n \log n + m \log n)$

$= O((n+m) \log n)$ if $m = \Theta(n^2)$

$= O(m \log n)$.

Proof of correctness By Induction



$\forall x \in S$: d[x] is length of correct shortest path from vertex s to x —(I)

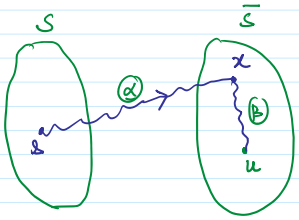
$\forall x \in \bar{S}$: d[x] is length of shortest path from s to x using only vertices of S —(II)

Let u is vertex of smallest label in $\bar{S}$. We shift u from $\bar{S}$ to S

Need to show that updated partition S+u & $\bar{S}/u$ also satisfies (I) & (II)

Claim (I) d[u] should be the shortest path from s to u, where u is the vertex with shortest label in set $\bar{S}$.

Proof Assume contrary: ∃ a shortest path from s to u via vertex $x \in \bar{S}$



Shortest path from x to u path $s \rightsquigarrow x + x \rightsquigarrow u$

Weight of entire path = weight of path $s \rightsquigarrow x$ + weight of path $x \rightsquigarrow u$ (non negative)

(i)

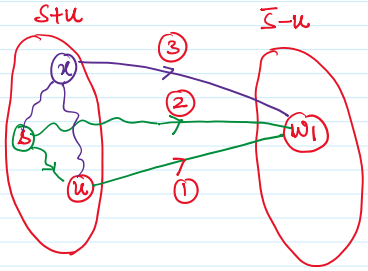
$\geq d[x]$

By assumption path (i) is shorter than d[u]

$\Rightarrow$ $d[x] \leq d[u]$

Implies that algo should have picked x instead of u

Contradiction !!



$d[w_1] \leftarrow \min \{ d[w_1], d[u] + w_t(u, w_1) \}$

There are three shortest path from s to w_1

(1) $s \rightsquigarrow u \rightarrow w_1$: shortest path from s to u and edge (u, w_1)

(2) Some other path from s to w_1

(3) Shortest Path from s to u and path from u to x and path from x to w_1

Path ① & ② are taken care by update rule $d[w_1] \leftarrow \min\{d[w_1], d[u] + w_t(u, w_1)\}$

Claim③

To complete the proof need to show that the length of path $s \rightsquigarrow u \rightsquigarrow x \rightsquigarrow w_1$ is more than $d[w_1]$

Claim② length of path $s \rightsquigarrow u \rightsquigarrow x \rightsquigarrow w_1$ is atleast length of path from $s \rightsquigarrow x \rightsquigarrow w_1$

Proof Note the vertex (x) was already in set S and vertex (w) just came in

$\Rightarrow$ Shortest path from $s \rightsquigarrow x$ doesn't need vertex (w)

weight of path $(s \rightsquigarrow u \rightsquigarrow x) \geq$ weight of path $(s \rightsquigarrow x)$

then Claim② is proved.

Proof of Claim③

weight of path $(s \rightsquigarrow u \rightsquigarrow x \rightarrow w_1) \geq$ weight of path $(s \rightsquigarrow x \rightarrow w_1) \geq d[w_1]$

By def.